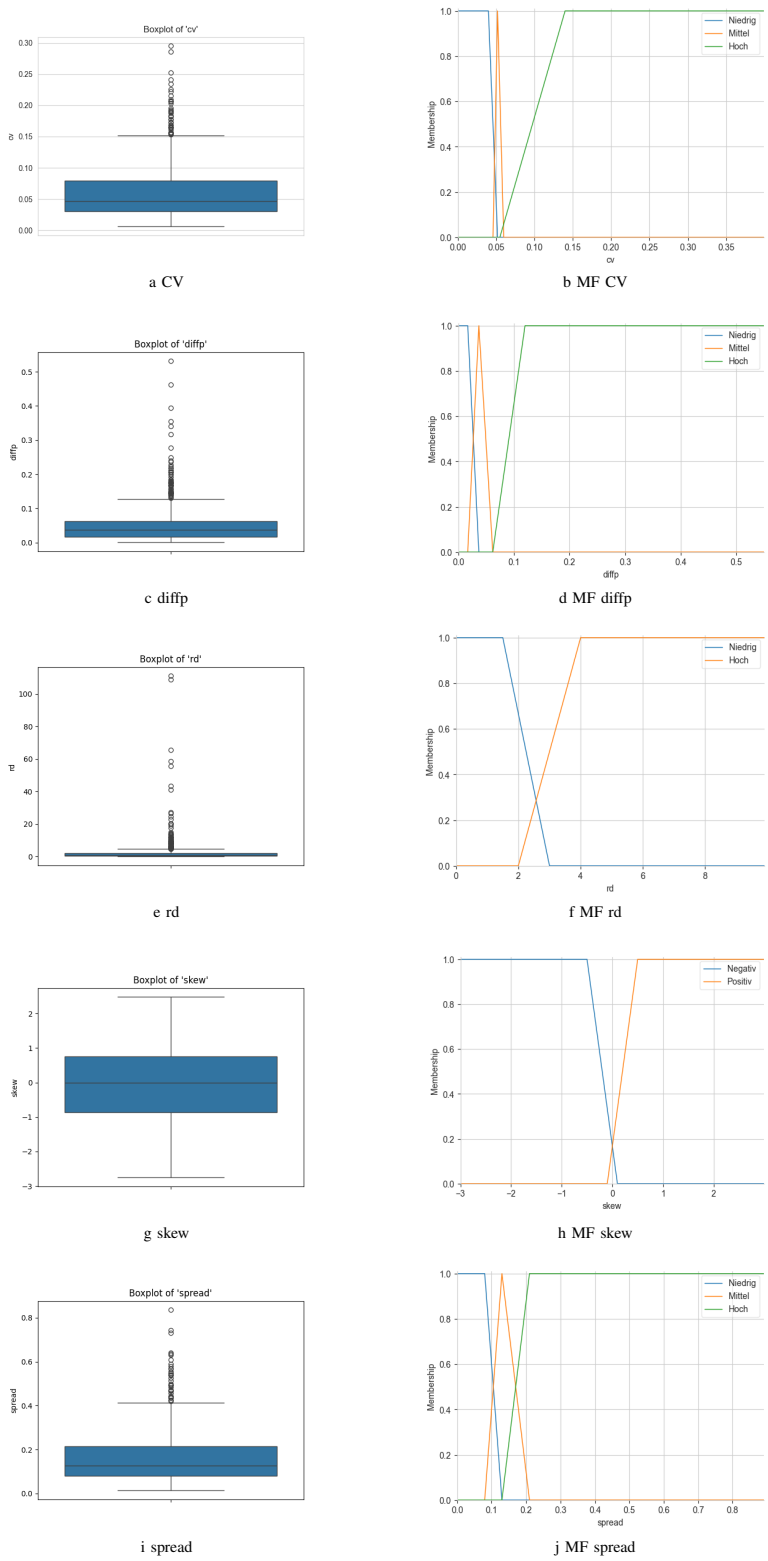
Supplementary Material for: Suspicion Screening: A Fuzzy-Logic Approach to Bid-Rigging Detection

I. Overview

This document serves as the supplementary material for the paper "Suspicion Screening: A Fuzzy-Logic Approach to Bid-Rigging Detection". It contains detailed visualizations of the input variable distributions (boxplots), the corresponding fuzzy membership functions, and the results of the fuzzy c-means clustering algorithm used to derive the data-driven rules.

Appendix A
Boxplots and Membership Functions

Figure 1 illustrates the statistical distribution of the input features (left) and the corresponding fuzzy membership functions (right).



a CV

b MF CV

c diffp

d MF diffp

e rd

f MF rd

g skew

h MF skew

i spread

j MF spread

Figure 1. Overview of Feature Distributions & Membership Functions.

Appendix B
Results Fuzzy C-Means Clustering

This appendix details the fuzzy c-means clustering and the centroids derived from the algorithm applied to the standardized training data. These centroids represent the archetypes used to generate the data-driven fuzzy rules.

A. Evaluation of Clustering Method

Table I presents a SWOT analysis of the chosen algorithm.

Table I
SWOT Analysis: Fuzzy C-Means vs. Advanced Clustering (e.g., Gustafson-Kessel)

Category	Analysis
Internal Factors (Strengths & Weaknesses)	
Strengths	• **Parsimony:** Minimizes model complexity to prove the viability of the Fuzzy approach itself without relying on complex tunings. • **Stability:** Robust against singularity issues that often plague covariance-based algorithms like Gustafson-Kessel on small datasets [1]. • **Baseline:** Established industry standard [2], serving as a reliable benchmark.
Weaknesses	• **Geometric Bias:** Constrained to spherical clusters (Euclidean distance). Fails to capture correlation-based (ellipsoidal) shapes [3]. • **Structure Mismatch:** Does not align well with linear dependencies required for optimal TSK-1 performance.
External Factors (Opportunities & Threats)	
Opportunities	• **Future Research:** Implementing shape-adaptive algorithms (Gustafson-Kessel) could improve TSK-1 accuracy by modeling linear substructures [1]. • **Noise Handling:** Possibilistic C-Means could reduce outlier impact in future iterations [4].
Threats	• **Singularity Risk:** Advanced methods (Gustafson-Kessel) are prone to numerical instability on limited datasets like the cartel cases used here. • **Overfitting:** Increasing clustering complexity increases the risk of modeling noise rather than economic patterns.

B. Cluster Determination and Centroids

Figure 2 shows the Elbow plot used to determine the optimal number of clusters. The resulting coordinates of the centroids are listed in Table II.

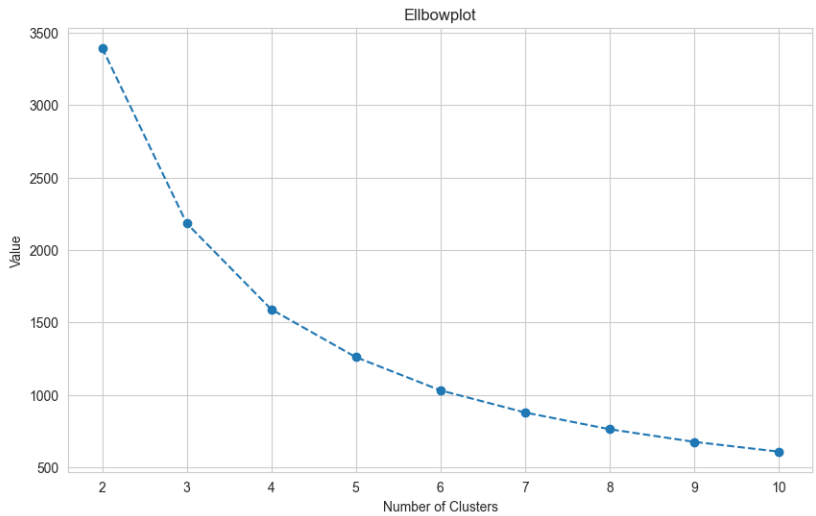


Figure 2. Elbow plot to determine the number of clusters.

Table II
COORDINATES OF THE CLUSTER CENTROIDS (STANDARDIZED)

Feature	**Centroid 1**	**Centroid 2**	**Centroid 3**
cv	-0.4228	0.7070	-0.4862
spread	-0.3401	0.6479	-0.4820
diffp	0.3118	0.0253	-0.3909
rd	0.5015	-0.2499	-0.1771
rdnorm	1.2188	-0.5773	-0.2237
rdalt	0.7894	-0.3435	-0.2323
skew	-1.0625	0.5527	0.1851
ks	1.1704	-0.5715	-0.1811
cartel	**0.5560**	**-0.8448**	**0.6269**

REFERENCES

[1] R. Babuška, *Fuzzy modeling for control*. Springer Science & Business Media, 1998.

[2] J. C. Bezdek, *Pattern recognition with fuzzy objective function algorithms*. Plenum Press, 1981.

[3] D. E. Gustafson and W. C. Kessel, "Fuzzy clustering with a fuzzy covariance matrix," in *1979 IEEE Conference on Decision and Control including the 18th Symposium on Adaptive Processes*. IEEE, 1979, pp. 761–766.

[4] R. Krishnapuram and J. M. Keller, "A possibilistic approach to clustering," *IEEE Transactions on Fuzzy Systems*, vol. 1, no. 2, pp. 98–110, 1993.